

7.3 The management of urban growth

By 2050, a predicted 66% of people will live in cities, yet urban areas will take up only 2% of the world's land surface area. This urban growth means significant challenges for planners and governments as they try to provide housing, transport, employment, energy, and services to support large numbers of people in high density urban areas. Urban growth needs to be sustainably managed.

Sustainable urban management refers to a city's use of resources and space in a way that meets the needs of its residents today without negatively impacting residents' needs in the future. Urban growth needs to be managed economically, socially, and environmentally to be truly sustainable. There are some interesting examples of sustainable management strategies of urban growth around the world.

Housing in Rio, Brazil

One challenge common to many cities in LICs is informal settlements. In Rio de Janeiro, Brazil, about one in five people live in informal settlements, or *favelas*. Different management strategies include government redevelopment (**top-down**) schemes. This approach includes techniques such as demolishing unplanned settlements and rebuilding on the land. A perhaps more successful strategy is **self-help** or **bottom-up schemes** that involve the residents of such settlements. This includes techniques such as providing residents with building materials, tools, and training which they use to construct and improve their existing homes.

The Favela Bairro Project aims to provide residents with essential services. The first phase of the project provided paved roads, water supply pipes, sewerage/drainage systems, creches, leisure facilities, and sports areas. The second phase aimed to generate employment opportunities and provide education to bring the *favela* dwellers into mainstream society and away from crime. Large-scale, top-down projects can be transformational but often run over budget. Bottom-up projects involve local people in the decision-making process but can be relatively small-scale.

TASK 1

- a What is the difference between 'top-down' and 'bottom-up' housing management schemes?
- b Outline the advantages and disadvantages of the Favela Bairro Project for:
 - i Favela residents
 - ii Urban planners
 - iii The government



Figure 7.5: *Favelas* in Rio de Janeiro, Brazil

Transport schemes

London, UK and congestion charges

An example of a controversial sustainable transport management strategy is London's **congestion charge scheme**. This is a charge that motor vehicle drivers pay when they enter the congestion charge zone. The current charge is £15 for every day you travel between 07:00 and 18:00. There is a penalty fine of between £65 and £195. In 2019, the Ultra Low Emission Zone (ULEZ) was introduced, which applies every hour of every day to vehicles that do not meet the emissions standards. In 2021, the ULEZ was expanded to cover the Inner London area. The congestion charge exemption does not apply to pure electric vehicles.

This strategy has both advantages and disadvantages for managing the impacts of traffic congestion in London (Table 7.4).



Figure 7.6: Sustainable transport management in London, UK

Table 7.4: The advantages and disadvantages of the London Congestion Charge

Advantages	Disadvantages
Bus journeys are quicker with fewer delays. Bus travel has increased by 33% since the charge was introduced.	The number of private for-hire vehicles (e.g. mini cabs and Ubers) has increased. This has reduced the speed of traffic through the city centre. (Taxis and minicabs are exempt from paying the congestion charge).
Reduces car vehicle emissions and improves air quality. Congestion has reduced by 30%, shortening travel times.	Lost business for some companies in the city centre as some people may choose not to pay the charge and instead to visit out of town shopping centres. This will lead to a decline in customers to city centres.
More journeys have switched to walking, cycling, and public transport (all of which have become safer).	It is an expensive scheme for the local government to run. It requires high technology to monitor, track, and collect payments.
Raises investment funds for London’s transport system.	The cost of the charge and of fines has seen a growth in false number plates and illegal minicab firms.

Reducing cars on the road in La Paz, Bolivia

Air quality is a major problem in many urban areas. In Bolivia, a cable-car system connects El Alto, the world’s highest city, and Bolivia’s second largest city with its capital, La Paz. It is the largest urban cable-car system in the world. The cable-car system offers an affordable, noise- and air pollution-free route for the 85% of the population that relies on public transport. On average 160 000 people a day pay just \$0.40 each to travel the 10 km stretch. It is an affordable solution, allowing poorer residents access to jobs and services away from their homes. It also reduces traffic congestion in the city and vehicle fuel costs.



Figure 7.7: Cable car system in La Paz, Bolivia

Innovative public transport in Curitiba, Brazil

Curitiba is an excellent model for sustainable urban development. The city experienced rapid population growth from 300 000 in 1950 to 3.86 million in 2023. However, it has avoided many of the problems associated with urban growth. Innovative planning in public transportation and improved pedestrian infrastructure has helped to reduce reliance on cars in urban areas, improve air quality, and reduce greenhouse gas emissions by around 40%. Urban planning techniques include:

- integrating public bus routes, the road network, and cycle paths into the urban life of the city
- improving public transport where there are local bus routes interlinked with faster express bus lanes
- buses have extra-wide doors to allow passengers on and off more quickly, and can be double and triple length to accommodate rush-hour passenger demand.

Over 70% of Curitiba's residents use the BRT (Bus Rapid Transit) system to commute to work every day. It is estimated that the transport system has reduced the need for 40 million car trips per year, saving 40 million litres of fuel every year. Importantly, it is cheap too. People in Curitiba spend only about 10% of their income on travel, which is well below Brazil's national average.



Figure 7.8: Bus in Curitiba: the tube-shaped bus stop is more efficient as passengers can enter and leave buses without steps and can pay fares at the stop rather than on the bus.

Developing green spaces

Green infrastructure such as green roofs and urban gardens are effective sustainable urban management strategies. ‘Green’ features improve air and water quality and provide green spaces for recreation and relaxation. This strategy is both environmentally and socially advantageous, with access to nature in cities seen as an important part of health and wellbeing for people living in urban areas.

261

In Mexico City, for example, *azoteas verdes* (green roofs) have become a solution to clean the city’s air. The green roofs produce oxygen and filter out some carbon dioxide. They also help regulate the temperatures of the offices below and soak up rainwater to keep buildings dry. They can also help shade and cool spaces in cities, reducing energy use for air conditioning. Another example is Bishan-Ang Mo Kio Park in Singapore, a former industrial area, which has been transformed into a wildlife park with natural water flow and wildflowers. Within the first two years of its existence, biodiversity naturally increased by 30%.



Figure 7.9: Green roofs in Mexico City, Mexico



TASK 2: Study Source **A**

- a** Study Source A. Identify five examples of sustainable urban management.
- b** Describe and explain one economic, one social, and one environmental management strategy you have identified.
- c** In pairs, research a city that has sustainable management strategies to tackle some of the challenges of rapid urban growth. Produce a poster to show, describe, and explain the strategies.
- d** Evaluate the strengths and weaknesses of these sustainable management strategies.

PROJECT

You have been hired to design a new sustainable city for India to help resolve some of the issues created by their rapidly growing urban population. Your city should consider strategies to manage social, economic, and environmental problems.

In small groups, you will give a pitch of your city to see if your teacher will choose your design. You can present your sustainable city as you choose. For example, you could present an annotated design sketch or an annotated design model. Your teacher will assess the geographical content of your final design and pitch, as well as the way in which you work together as a group throughout the process.

You should consider:

- **social solutions:** the city should be able to cope with the large numbers of people and provide its population with housing, education, and healthcare
- **environmental solutions:** the city should provide solutions to pollution and meet demand for energy, food, and water
- **economic solutions:** the city should be economically self-sustaining and an affordable place for people to live.

REFLECTION

Evaluate how you worked as a group during this project:

- what were your group strengths and weaknesses?
- what were your individual strengths and weaknesses?
- how could you improve your group work skills?
- how could you improve your individual contribution to group work?